

A comparative study of compression behaviors of cross-laminated timber and glued-laminated timber columns

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HIGHLIGHTS

- GLT column specimens had a greater compressive loading capacity than CLT column specimens.
- CLT column specimens had better ductility and energy absorption than GLT column specimens owing to the cross-lamination.
- The stress-strain relationship of CLT and GLT columns had three stages and can be modeled.
- Failure modes of GLT column specimens were simple but they were relatively complicated for CLT column specimens.

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ABSTRACT

This work investigated the different axial compression behaviors of cross-laminated timber columns (CLTCs) and control glued-laminated timber columns (GLTCs). The average compression modulus of elasticity (MOE) and strength values of the CLTCs were lower than those of the control GLTCs and appeared to be dependent on the area ratio in the axial direction. The column length had different effects on the compression behaviors of CLTCs and GLTCs, and compared with the GLTCs, the CLTCs exhibited enhanced ductility and energy absorption properties due to the cross-lamination. Of the three models developed in this work, model I is recommended for GLTCs and model II for CLTCs.

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1. Introduction

At present, mass timber products (MTPs), including mainly structural composite lumber (SCL), glued-laminated timber (GLT, which is also called glulam), cross-laminated timber (CLT), and nailed-laminated timber (NLT), are increasingly used for building applications [1]. MTPs can be prefabricated with precise dimensions and openings in a factory, which enables faster assembly with minimal construction waste [1]. The key advantages of MTPs include their light weight, high integrity, fast and easy on-site installation, excellent thermal and acoustic insulation, good

durability, and quick investment return. With the advent of MTPs, the market for mass timber construction has emerged throughout the world over the past several years [2].

Among the MTPs, GLT and CLT are increasingly substituting solid timber as primary supporting structures [3]. GLT, which has a long history in developed countries, is a traditional structural engineered wood product comprising a number of laminates of dimension lumber bonded together with durable, moisture-resistant structural adhesives [4,5]. The grains of all laminations run parallel to the length of the member. GLT can be used in horizontal applications as a beam, in vertical applications as a column, or as curved, arched shapes, and GLT is available in a range of strength classes [6]. During GLT manufacturing, the key is to clamp the glued lay-up under lateral compression for a period of several hours to longer than one week. The ultimate dimensions of GLT can

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reach 14,000 mm (length) $\times$ 1200 mm (width) $\times$ 284 mm (depth). CLT is the latest addition to the MTP family, which is generally made by gluing adjacent layers of dimension lumber at 90° to each other to form a large-dimension panel [7]. CLT offers unlimited size and a potential for larger, taller, more energy efficient and durable building systems. The species used for GLT and CLT manufacturing are generally the same: mainly softwoods, including Norway spruce (*Picea abies*), white fir (*Abies alba*), Scots pine (*Pinus sylvestris*), European larch (*Larix decidua*), Douglas fir (*Pseudotsuga menziesii*), and western larch (*Larix occidentalis*) [8]. Spruce-pine-fir (SPF) and Norway spruce are the most common species for GLT and CLT manufacturing, respectively, in North America and Europe [7,9]. Other species including hardwoods can also be used for CLT [9–14]. This large variation in wood selection is possible because of the solid composition and homogenization of panel properties within CLT.

According to current manufacturing methods [15], CLT panels must be formed in a large-scale press under pressure in the vertical and horizontal directions. When polyurethane (PUR) is used, a 1–2 h period of cold press is necessary. Obviously, the manufacturing efficiency for CLT panels is higher than that for GLT panels. From an application standpoint, CLT is primarily used for floors, walls and roofs and is processed less often for beam and column members. By contrast, GLT is mostly used for beam and column applications. The use of bolted GLT connections with slotted-in steel plates in mass timber constructions imparts GLT with particular advantages with respect to assembly and aesthetics. However, bolted GLT connections have low moment resistances; thus, unexpected brittle failures often occur because of low tensile strength perpendicular to the wood grain, particularly when the connection is subjected to a large rotational displacement. According to a previous work [16], brittle splitting cracks in the perpendicular-to-grain direction were the main cause of premature failures in beam-to-column glulam connections under monotonic and reversed cyclic loading. Through reasonable perpendicular alignment of adjacent layers of dimension lumber, CLT members in the perpendicular-to-grain direction could exhibit greater strength than GLT members and avoid brittle failures [17]. This concept was used by Blaß et al. in bolted glulam butt joint connections [18] and led to excellent ductility compared with that of unreinforced or self-tapping-screw-reinforced connections. Wang et al. obtained a better ductile failure mode by using the cross-laminated method in the node region, which was found to dissipate 25% more energy than GLT connections [19]. In addition, Ma et al. studied the relationship between dowel orientation and wood grain orientation with respect to the compressive failure modes of Douglas fir GLT specimens. Their results illustrated that a dowel hole set in the parallel-to-grain direction of GLT is preferable to a dowel hole set in the perpendicular-to-grain direction [20]. Schickhofer et al. [21] comparatively studied the nail-holding performance of GLT and CLT and found that CLT had better ductility [22]. Thus, CLT could have better prospects than GLT for beam and column members in consideration of their better connections, as stated in the revised Euro-code 5-1-1 from 2018. [23].

Some scholars [24–26] carried out detailed research on the out-of-plane compression performance of CLT specimens, but few scholars have studied the CLT under in-plane loading. To explore the performance of the CLT as an in-plane force member, Pang and Jeong studied the effects of the load distribution and the weakest layer under in-plane compression loading and proposed a new prediction method [27]. However, an in-depth study of CLT column components has not yet been carried out.

At present, forest resources are changing worldwide over time from old growth to second growth with an increasing volume of shorter rotation plantations [28–30]. Western hemlock (*Tsuga heterophylla* (Raf.) Sarg) presents the largest component of

second-growth forests in the coastal forest region of British Columbia, Canada. The current applications of western hemlock are limited to products such as solid wood, pulp and paper. Therefore, the value of western hemlock has not been fully recovered [31]. Based on the characteristics of western hemlock, its conversion into value-added products for new building applications. Such products should include wall, floor and roof panels, as well as beams and columns. Wang et al. (2018) studied the bond quality and durability of Canadian western hemlock CLT and identified its potential in the manufacture CLT [32]. He et al. (2018) investigated the out-of-plane bending properties and compression properties of hemlock CLT panels [33]. Although the aforementioned studies provided a basis for wall, floor and roof applications of hemlock CLT, the compression behavior of hemlock CLT used as a column has not been studied.

The objective of this work was to investigate the different axial compression behaviors of CLT and control GLT. Based on their compressive stress-strain relationships, a theoretical compression model was thus developed to quantify their axial compression behaviors.

2. Materials and methods

2.1. Specimen preparation

This study was conducted at the pilot plant of Ningbo Sino-Canada Low-Carbon Technology Research Institute, P. R. China. The No. 2 & Btr 14-ft (4.3 m) heat-treated (HT) & kiln-dried (KD) western hemlock 2 $\times$ 6 dimension lumber with a target moisture content (MC) of 19% or less was imported from Canada. The lumber pieces were then conditioned to achieve an MC of 9–15%. An HM 200 log acoustic tester was used to preselect lumber pieces with average sound transmission speeds of 4.0–4.5 m/s for CLT panel manufacturing to minimize property variation. Approximately 120 lumber pieces were selected and then planed to a thickness of 35 mm before gluing. A single-component PUR was used to glue the hemlock lumber. To help identify the different compression behaviors of the CLT and control GLT, the manufacturing parameters were exactly the same for both products: an opening time of 20 min, a glue application rate of 160 g/m², an applied pressure of 1.0 MPa, and a pressing time of 1 h. Twenty-seven (27) lumber pieces were used for each 3-ply CLT or GLT panel. Two replicates were used, which consumed a total of 108 lumber pieces. The 3-ply CLT and control GLT panels were manufactured with dimensions of 4300 mm $\times$ 1200 mm $\times$ 105 mm. After edge-trimming and cutting, the final dimensions of the panels were 4200 mm $\times$ 1100 mm $\times$ 105 mm. The configurations of the CLT and control GLT panels are shown in Fig. 1a and b, respectively.

The CLT and GLT panels were further cut into two square column specimens according to Fig. 1c: CLT column (CLTC) and GLT column (GLTC) specimens. The wood column specimens that were 210 mm long were denoted as CLTCa and GLTCa, and those 420 mm long were denoted as CLTCb and GLTCb (Fig. 1c).

2.2. Compression experiments and loading scheme

The compression performances of the CLTC and GLTC specimens were studied in this work. To test the compressive modulus of elasticity (MOE) and Poisson's ratio and reduce system error resulting from inaccurate machining of the specimens, two strain gages were longitudinally and transversely affixed to each specimen face. Herein, the longitudinal direction indicates the length direction of the column, and the transverse direction indicates the perpendicular-to-length direction of the column. For the narrow face of a CLTC, the longitudinal strain gage was positioned at the

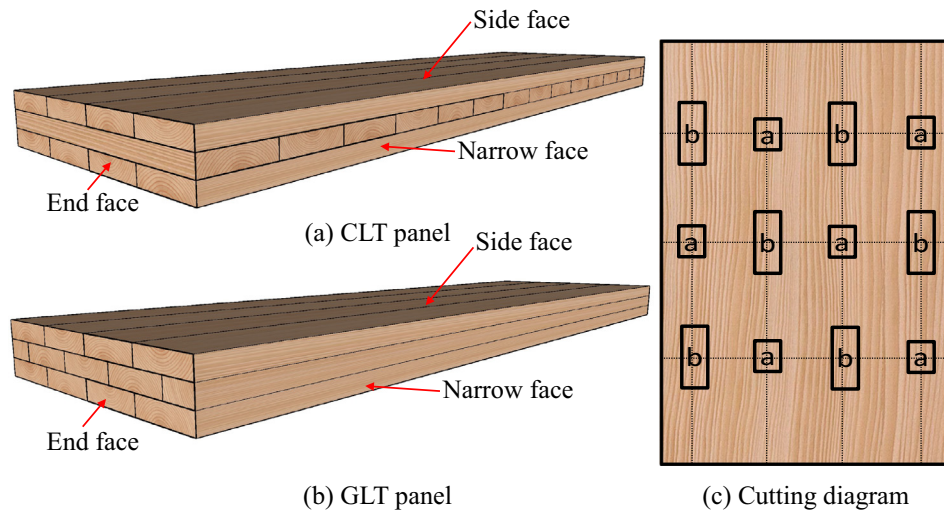


Fig. 1. CLT and GLT panel configurations and cutting diagram.

interlaminar gap, which formed an “L” with the transverse strain gage (Fig. 2a). However, the strain gages affixed on the side face of a CLTC were perpendicular to each other and formed a “T”, which is the same orientation as the strain gages on the side and narrow faces of the GLTC specimens (Fig. 2b). To appropriately match the strain gage size to the specimen size, strain gages with dimensions of 50 mm (length) $\times$ 5 mm (width) were selected for type “a” specimens (210 mm long). For the type “b” specimens (420 mm long), the effective sizes of the longitudinal and transverse strain gages were 100 mm $\times$ 5 mm (SZ120-100AA/DX, sensitivity coefficient $2.05 \pm 0.23\%$ and electric resistance $120 \pm 0.2\% \Omega$) and 50 mm $\times$ 5 mm (SZ120-50AA/DX, sensitivity coefficient $2.018 \pm 0.10\%$ and electric resistance $120 \pm 0.2\% \Omega$), respectively. All of the strain gages used in this work were purchased from Jinli Sensing Component Co. Ltd. (Xingtai, Hebei Province, China).

The test setup is illustrated in Fig. 2c. The displacement of the specimen along the axial direction was measured using two symmetrically located displacement sensors with a range of ± 25 mm and a correction coefficient of $0.005 \text{ mm}/\mu\epsilon$ (YHD-50, Liyang City Instrument and Meter Plant, China). The column specimens were axially compressed using a universal testing machine (SANS, MTS

Co. Ltd., USA) with a capacity of 2000 kN. The bottom plate of the loading machine was a spherical hinged-support, and the top plate of the loading machine was fixed-support. A static data acquisition instrument with a highest resolution of $0.1 \mu\epsilon$ and a sampling frequency of 2500 Hz (TDS-530, Tokyo Institute of Testing, Japan) was used to simultaneously collect the strain, load and displacement information. Notably, the top and bottom faces of the column specimens needed to be cut horizontally. If the specimens were not horizontally leveled, it would be necessary to lay standard sand not exceeding 3 mm in thickness on the top faces of the column specimens before testing. To better describe the failure modes, each side face of each specimen was denoted by a capital letter (A, B, C, and D). Face A was marked on one narrow face of the CLTC or GLTC specimen, while B, C, and D were sequentially assigned in a clockwise direction around the central axis.

The loading scheme was performed in accordance with GB/T 50329-2012 [34] and ASTM D198-15 [35]. The specimens were preloaded to check the testing machine and eliminate the gaps between the specimen and the plate or support. During the initial stage of formal loading, the load control mode was selected, and when the load was linearly increased to 80% of the estimated ultimate load, the control mode was changed to displacement control until the specimen was damaged. Fig. 3 shows the specific loading schemes used for the CLTC and GLTC specimens. Specifically, for the CLTC specimens, the preload was linearly increased to 55 kN at a rate of 1.0 kN/s and then linearly decreased to 25 kN at the same rate. The load was linearly cycled between 25 kN and 60 kN a total of five times to accurately evaluate the elastic modulus, and the load was then finally reduced to zero. During the formal loading stage of each CLTC, the load was increased from zero to 210 kN through load control at a rate of 1.0 kN/s and then increased through displacement control at a rate of 0.5 mm/min until the specimen was damaged. Similarly, for the GLTC specimens, the preload was linearly increased to 80 kN at a rate of 1.0 kN/s and then linearly reduced to 40 kN at the same rate. The load was linearly cycled between 40 kN and 80 kN for a total of five times to accurately evaluate the elastic modulus; the load was then finally reduced to zero. During the formal loading stage of each GLTC, the load was increased from zero to 340 kN through load control at the rate of 1.0 kN/s and then increased through displacement control at a rate of 0.5 mm/min until the specimen was damaged.

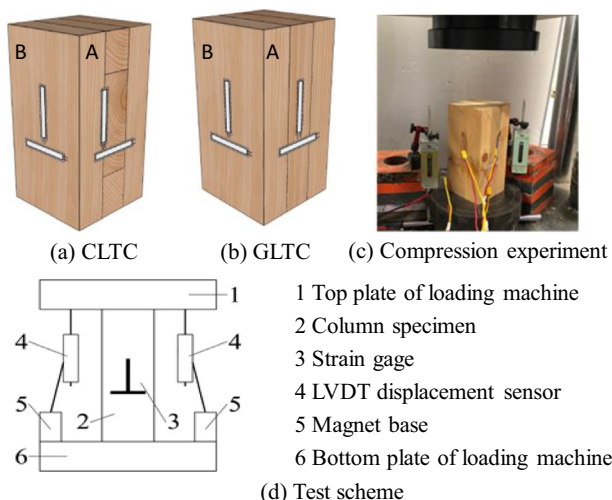


Fig. 2. Column specimens and test setup.

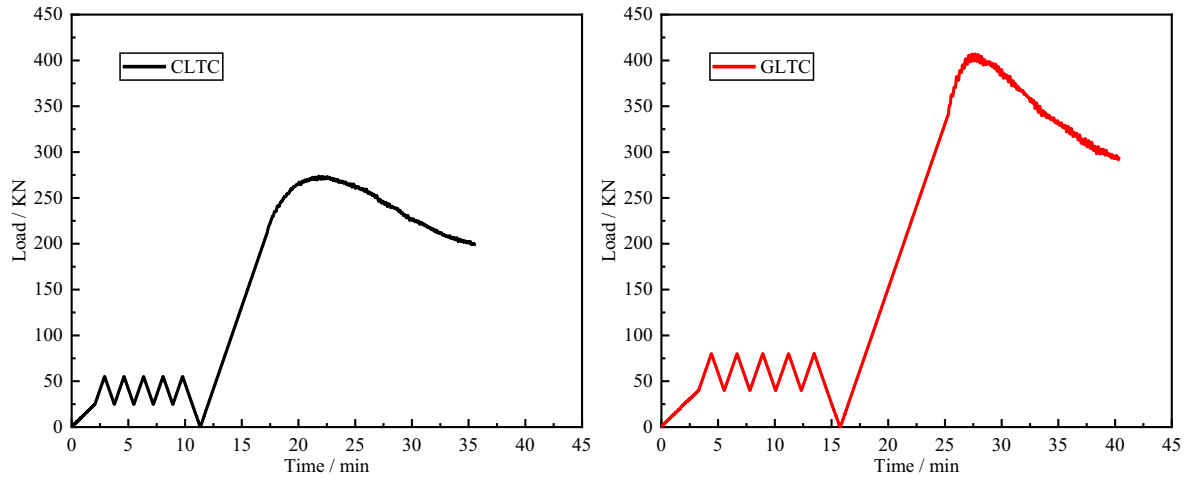


Fig. 3. Loading schemes for the CLTC and GLTC specimens.

3. Results and analysis

3.1. Compression behaviors of the two types of wood columns

The axial compression strength of a wood column (σ_{cu}) can be determined by the ultimate bearing capacity and cross-sectional area of the specimen, and the axial compression strength can be calculated by the following equation:

$$\sigma_{cu} = \frac{F_{cu}}{bh} \quad (1)$$

where b is the width of the specimen (mm), h is the depth of the specimen (mm), and F_{cu} is the ultimate bearing capacity (kN).

The compression MOE of the column was determined by the relationship between stress and strain during the elastic range, which can be calculated as follows:

$$E_i = \frac{\Delta\sigma_i}{\Delta\varepsilon_i} = \frac{\Delta F_i}{bh\Delta\varepsilon_i}, \quad E = \frac{\sum_{i=1}^n E_i}{n} \quad (2)$$

where E_i is the secant MOE in the i^{th} time interval, $\Delta\sigma_i$ and $\Delta\varepsilon_i$ are the incremental stress and strain in the i^{th} time interval of the elastic range, and ΔF_i is the incremental load in the i^{th} time interval of the elastic range, respectively.

Based on the longitudinal and transverse strains, the compressive Poisson's ratio, μ , which is defined as the transverse deformation resulting from the longitudinal deformation, can be calculated by the following equation:

$$\mu_i = -\frac{\Delta\varepsilon_{xi}}{\Delta\varepsilon_{yi}}, \quad \mu = \frac{\sum_{i=1}^n \mu_i}{n} \quad (3)$$

where μ_i is the Poisson's ratio in the i^{th} time interval of the elastic range and $\Delta\varepsilon_{xi}$ and $\Delta\varepsilon_{yi}$ are, respectively, the incremental transversal and longitudinal strains in the i^{th} time interval of the elastic range.

The axial compression strength and MOE of each column are listed in Table 1. The axial compression strength and MOE values of the CLTC specimens were clearly lower than those of the GLTC specimens, which is caused by the different lay-up structures of the CLT and GLT—specifically, the parallel-to-grain lumber arrangement of adjacent layers in GLT and the perpendicular-to-grain lumber arrangement of adjacent layers in CLT. In a sense, the core layer of a CLTC mainly connects two longitudinal layers and offers limited compressive capacity.

On average, the compression MOE and strength values of the CLTC specimens were approximately 39.4% and 34.6% lower than those of the control GLTC specimens, respectively. For the compression strength, the COV values of the CLTC specimens were greater than those of the control GLTC specimens because of the cross-lamination of the CLTC specimens. Moreover, the compression performance of the shorter column specimens varied more than that of the longer column specimens in each test group; this result might occur because the shorter column specimens are more likely influenced by the press plate according to the Saint-Venant principle. Compared with the shorter column, the longer column showed an increase in the compression strength for both GLTC and CLTC; however, the magnitude was much higher with GLTC (7.69%) than with CLTC (only 1.46%). Furthermore, the longer column resulted in an increase of 3.39% for the GLTC MOE but a decrease of 4.02% for the CLTC MOE.

Statistically, the effects of the column length on the compression strength of the two types of wood column specimens were analyzed by a one-way analysis of variance (ANOVA) (Table 2). The results show that the column length did not have a significant effect on the compression strength of the CLTC ($p = 0.697 > 0.05$) but had a significant effect on the compression strength of the GLTC ($p = 0.001 < 0.05$). The column length did not have significant effect on the compression MOE for either CLTC or GLTC ($p = 0.452 > 0.05$ and $p = 0.340 > 0.05$, respectively).

The absolute values of the Poisson's ratios of the CLTC and GLTC specimens are listed in Table 3. The absolute value of the Poisson's ratio that characterizes the radial deformation caused by a tangential compression is 0.382, which is lower than that for radial compression caused by a longitudinal deformation (0.485) according to the wood handbook [36]. The absolute value of the Poisson's ratio of the GLTC specimen on the narrow face had a mean of 0.513, which is larger than the relevant Poisson's ratio (0.485) of solid western hemlock wood; this phenomenon could result from microcracks that remained between layers of the lumber after gluing. However, the deformation characteristics of the GLTC specimen on the wide face were similar to those of western hemlock wood.

For the CLTC specimen, the absolute values of the Poisson's ratio on the narrow and wide faces were larger than those of solid western hemlock wood because of cross-lamination. The wood column specimens with cross-lamination had larger internal stresses resulting from the nonuniform deformations of the lumber in different directions under axial compression loading, which increases the transverse deformation of the narrow face of the CLTC

Table 1

Comparison of the axial compression performance between CLTC and GLTC.

Column type [*]	Compression strength		MOE	
	Mean (MPa)	Coefficient of variation (COV)	Mean (MPa)	Coefficient of variation (COV)
CLTCa ^{**}	23.92	7.22%	7255.90	6.08%
CLTCb ^{**}	24.27	5.14%	6964.37	5.46%
Mean	24.10	6.18%	7110.14	5.77%
GLTCa ^{**}	35.50	3.14%	11530.17	11.89%
GLTCb ^{**}	38.23	1.98%	11921.00	4.31%
Mean	36.87	2.56%	11725.59	8.10%

^{*} The total number of each type of column specimen is thirty-six (36).^{**} a represents the column with a length of 210 mm, and b represents the column with a length of 420 mm.**Table 2**

Variance analyses of the column length and compression performance of the CLTC and GLTC.

Column type	Degrees of freedom (DOF)	Compression strength			MOE		
		Standard deviation (SD)	F	p	SD	F	p
CLTCa [*]	11	1.727	0.160	0.697	862.58	0.611	0.452
CLTCb [*]		1.247			300.42		
GLTCa [*]	11	1.115	24.684	0.001	701.08	1.002	0.340
GLTCb [*]		0.758			650.36		

^{*} a represents the column with a length of 210 mm, and b represents the column with a length of 420 mm.**Table 3**

Absolute values of the Poisson's ratio of the CLTC and GLTC.

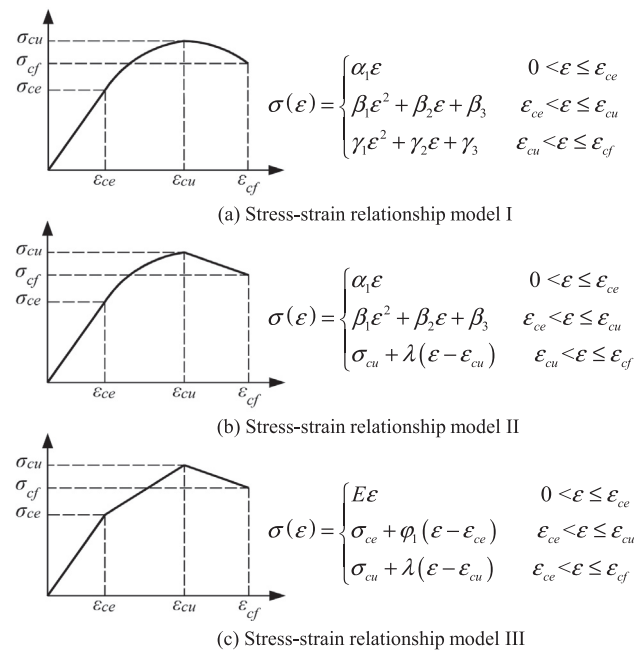
Column types	Narrow face		Wide face	
	Mean	COV	Mean	COV
GLTC	0.513	13.78%	0.469	14.20%
CLTC	0.753	40.20%	0.576	38.76%

specimen. The narrow and wide faces of the CLTC specimens had a much larger variation of the Poisson's ratio than those of the GLTC specimens. Those results also demonstrate that cross-lamination increases the variation of the Poisson's ratio of the wood columns.

3.2. Development of an axial compression stress-strain model

Based on previous publications [37–40], for the compression of the wood column specimens, this work used a linear function to describe the elastic range and either a linear function or a quadratic function to describe the elastoplastic range and the plastic range. As shown in Fig. 4, the stress-strain relationship between GLT and CLT was described by three different models. Fig. 4a depicts stress-strain model I, in which the elastic range is represented by a linear function and the elastoplastic and plastic ranges are represented by two different quadratic functions. Fig. 4b shows stress-strain model II, in which the elastic range is represented by a linear function, the elastoplastic range is represented by a quadratic function, and the plastic range is represented by a linear function. Fig. 4c shows stress-strain model III, wherein the elastic, elastoplastic and plastic ranges are all represented by linear functions. Note that 60% of the ultimate stress of each GLTC and CLTC specimen was defined as the proportional ultimate stress. The GLTC specimen was considered damaged when it reached its ultimate stress and the load subsequently decreased to 80% of the ultimate stress [41]. However, for the CLTC specimen, this proportion was 90%.

In these models, ε_{cu} is the ultimate strain corresponding to the ultimate stress σ_{cu} of the test piece; ε_{ce} is the strain value corresponding to the ultimate strength σ_{ce} of the specimen, which is generally defined as 0.45–0.60 ε_{cu} [42] and ε_{ce} is defined as 0.60 ε_{cu} in this work; ε_{cf} is the failure strain corresponding to the failure stress σ_{cf} ; and α_1 , β_1 , β_2 , β_3 , γ_1 , γ_2 , and γ_3 are coefficients in the

**Fig. 4.** Three models of the axial compression behaviors of the wood column specimens.

stress-strain model. Each coefficient in the stress-strain model was obtained according to the boundary conditions of the test piece within each range. The boundary conditions of the elastic range were the proportional ultimate stress and strain values of the specimen at the proportional limit, the boundary conditions of the elastoplastic range were the ultimate stress and strain values of the specimen under the ultimate load, and the boundary conditions of the plastic range were the failure stress and strain values of the specimen when it was destroyed.

For model I, based on Hooke's law, the elastic range stress-strain relationship can be expressed as follows:

$$\frac{d\sigma}{d\varepsilon} = E = \alpha_1 \quad (4)$$

The elastoplastic range of model I went through two points, $(\varepsilon_{ce}, \sigma_{ce})$ and $(\varepsilon_{cu}, \sigma_{cu})$; thus,

$$\sigma_{ce} = \beta_1 \varepsilon_{ce}^2 + \beta_2 \varepsilon_{ce} + \beta_3, \sigma_{cu} = \beta_1 \varepsilon_{cu}^2 + \beta_2 \varepsilon_{cu} + \beta_3 \quad (5)$$

Moreover, while progressing to the point $(\varepsilon_{cu}, \sigma_{cu})$, the stress reached its maximum value. Thus,

$$\left. \frac{d\sigma}{d\varepsilon} \right|_{\varepsilon=\varepsilon_{cu}} = 2\beta_1 \varepsilon_{cu} + \beta_2 = 0 \quad (6)$$

In addition,

$$\begin{aligned} \beta_1 &= \frac{\sigma_{ce} - \sigma_{cu}}{(\varepsilon_{ce} - \varepsilon_{cu})^2}, \beta_2 = \frac{-2\varepsilon_{cu}(\sigma_{ce} - \sigma_{cu})}{(\varepsilon_{ce} - \varepsilon_{cu})^2} \text{ and} \\ \beta_3 &= \frac{\varepsilon_{ce}^2 \sigma_{cu} - 2\varepsilon_{ce} \varepsilon_{cu} \sigma_{cu} + \varepsilon_{cu}^2 \sigma_{ce}}{(\varepsilon_{ce} - \varepsilon_{cu})^2} \end{aligned} \quad (7)$$

Because the plastic phase passes two points $(\varepsilon_{cu}, \sigma_{cu})$ and $(\varepsilon_{cf}, \sigma_{cf})$, the ultimate stress can be expressed as

$$\sigma_{cu} = \gamma_1 \varepsilon_{cu}^2 + \gamma_2 \varepsilon_{cu} + \gamma_3, \sigma_{cf} = \gamma_1 \varepsilon_{cf}^2 + \gamma_2 \varepsilon_{cf} + \gamma_3 \quad (8)$$

Moreover, the maximum value is obtained at point $(\varepsilon_{cu}, \sigma_{cu})$; therefore,

$$\left. \frac{d\sigma}{d\varepsilon} \right|_{\varepsilon=\varepsilon_{cu}} = 2\gamma_1 \varepsilon_{cu} + \gamma_2 = 0 \quad (9)$$

In addition,

$$\begin{aligned} \gamma_1 &= -\frac{\sigma_{cu} - \sigma_{cf}}{(\varepsilon_{cu} - \varepsilon_{cf})^2}, \gamma_2 = \frac{2\varepsilon_{cu}(\sigma_{cu} - \sigma_{cf})}{(\varepsilon_{cu} - \varepsilon_{cf})^2} \text{ and} \\ \gamma_3 &= \frac{\varepsilon_{cu}^2 \sigma_{cf} - 2\varepsilon_{cu} \varepsilon_{cf} \sigma_{cu} + \varepsilon_{cf}^2 \sigma_{cu}}{(\varepsilon_{cu} - \varepsilon_{cf})^2} \end{aligned} \quad (10)$$

For model II, the boundary conditions were used to obtain the following expression:

$$\lambda = \frac{\sigma_{cf} - \sigma_{cu}}{\varepsilon_{cf} - \varepsilon_{cu}} \quad (11)$$

Additionally, for model III, the boundary conditions were used to obtain the following expression:

$$\varphi_1 = \frac{\sigma_{cu} - \sigma_{ce}}{\varepsilon_{cu} - \varepsilon_{ce}} \quad (12)$$

In these stress-strain relationship models, σ_{cu} , σ_{ce} , and σ_{cf} are the mean values of the axial compression characteristic constants of the specimens. The axial compression characteristic constants of the CLTC and GLTC specimens are shown in Tables 4 and 5, respectively. The stress-strain model was calibrated by substituting the GLTC and CLTC characteristic constants into the stress-strain models I, II, and III. The results are shown in Figs. 5–7.

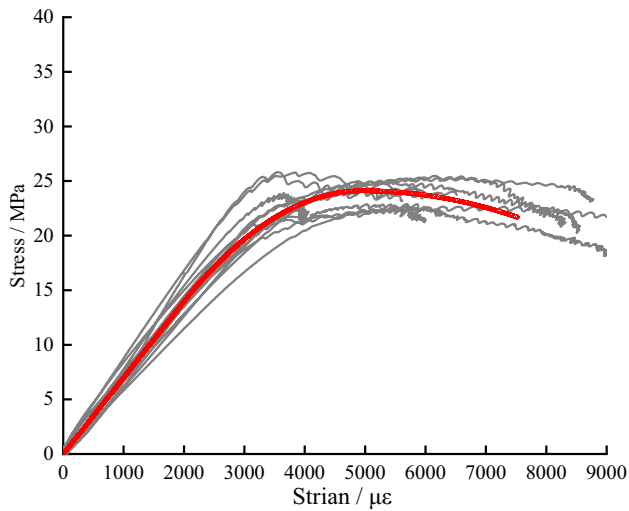
Figs. 5 through 7 show that the models can well reflect the stress-strain relationship of the CLTC and GLTC specimens during axial compression. The axial compressive stress of the GLTC specimens decreased sharply when the stress reached the ultimate stress, whereas the stress of the CLTC specimens decreased slowly; thus, the CLTC specimens exhibited good ductility and energy absorption performance. In contrast, model I used two different quadratic functions for the elastoplastic range and the plastic range, which agreed best with the actual stress-strain curves of the CLTC and GLTC specimens. Fig. 6 shows that model II differed from model I in that the plastic range changed from a quadratic function to a linear function. Although the form was simplified, the theoretical and actual stress-strain relationships of the CLTC and GLTC specimens were still well matched, especially for the CLTC specimens. In Fig. 7, the elastic, elastoplastic and plastic ranges of stress-strain model III adopted linear functions, and the

Table 4
Axial compression characteristic constants of the CLTC specimens.

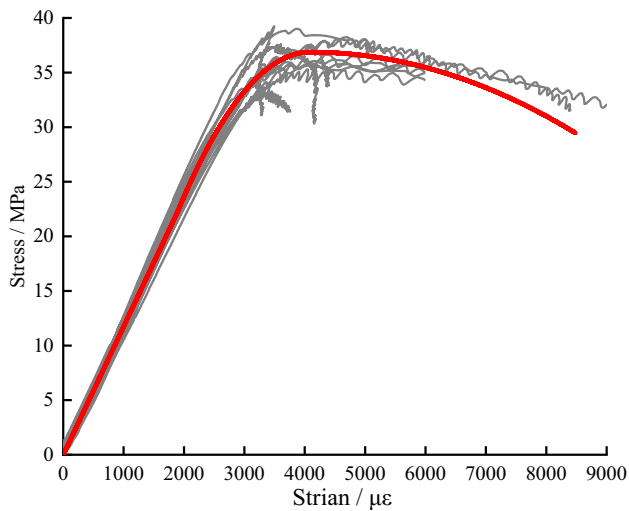
Specimen number	Ultimate stress σ_{cu} (MPa)	COV (%)	Ultimate strain ε_{cu} (10^{-6})	Proportionallimit stress σ_{ce} (MPa)	Proportional limit strain ε_{ce} (10^{-6})	Failure strain σ_{cf} (MPa)	Failure stress ε_{cf} (10^{-6})	Mean
CLTCa-1	25.80	7.17%	3563.00	15.48	1912.15	23.22	5593.78	$\varepsilon_{cu} = 4962.88$
CLTCa-2	23.87		3656.00	14.32	1850.46	21.48	5483.24	$\sigma_{cu} = 24.10$
CLTCa-3	24.13		4979.33	14.48	2117.88	21.72	10371.82	$\varepsilon_{ce} = 2069.28$
CLTCa-4	21.50		3517.00	12.90	1670.32	19.35	–	$\sigma_{ce} = 14.46$
CLTCa-5	25.77		3849.50	15.46	1828.52	23.19	6257.41	$\varepsilon_{cf} = 7518.99$
CLTCa-6	22.48		5623.67	13.49	2371.00	20.23	5510.48	$\sigma_{cf} = 21.69$
CLTCb-1	22.86		5028.00	13.71	2037.12	20.57	7708.28	
CLTCb-2	22.65		6301.67	13.59	1964.56	20.38	11494.17	
CLTCb-3	25.06		5380.67	15.03	2244.37	22.55	7893.14	
CLTCb-4	25.32		5966.75	15.19	2115.09	22.79	7215.95	
CLTCb-5	24.28		5514.50	14.57	2298.54	21.85	–	
CLTCb-6	25.47		6174.50	15.28	2421.35	22.92	7661.67	

Table 5
Axial compression characteristic constants of the GLTC specimens.

Specimen number	Ultimate stress σ_{cu} (MPa)	COV (%)	Ultimate strain ε_{cu} (10^{-6})	Proportional limit stress σ_{ce} (MPa)	Proportional limit strain ε_{ce} (10^{-6})	Failure strain σ_{fc} (MPa)	Failure stress ε_{fc} (10^{-6})	Mean
GLTCa-1	35.80	4.59%	4094.33	21.48	1708.18	28.64	–	$\varepsilon_{cu} = 4087.44$
GLTCa-2	35.70		4138.25	21.42	1891.85	28.56	8789.49	$\sigma_{cu} = 36.87$
GLTCa-3	35.28		4036.50	21.17	1870.56	28.22	16620.16	$\varepsilon_{ce} = 1881.40$
GLTCa-4	33.42		3412.00	20.05	1759.61	26.73	3990.00	$\sigma_{ce} = 22.12$
GLTCa-5	36.62		4371.33	21.97	2026.18	29.29	9527.45	$\varepsilon_{cf} = 8474.87$
GLTCa-6	36.17		4468.67	21.70	1886.58	28.94	14931.73	$\sigma_{cf} = 29.49$
GLTCb-1	37.30		4087.00	22.38	1882.20	29.84	4149.38	
GLTCb-2	39.00		3873.25	23.40	1823.74	31.20	–	
GLTCb-3	37.58		3606.00	22.55	1825.19	30.06	4174.65	
GLTCb-4	39.21		3493.50	23.53	1939.51	31.37	3283.72	
GLTCb-5	38.24		4699.25	22.94	1977.05	30.59	8355.23	
GLTCb-6	38.07		4769.25	22.84	1986.16	30.45	10926.89	



(a) Model I of the CLTC specimens



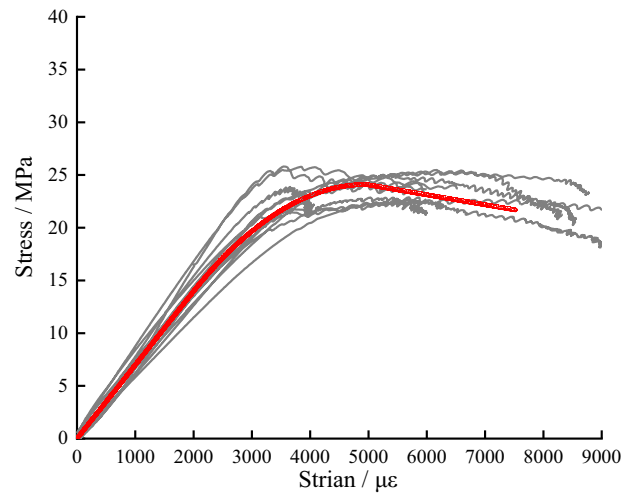
(b) Model I of the GLTC specimens

Fig. 5. Model I of the axial compression stress-strain relationships of the CLTC and GLTC specimens.

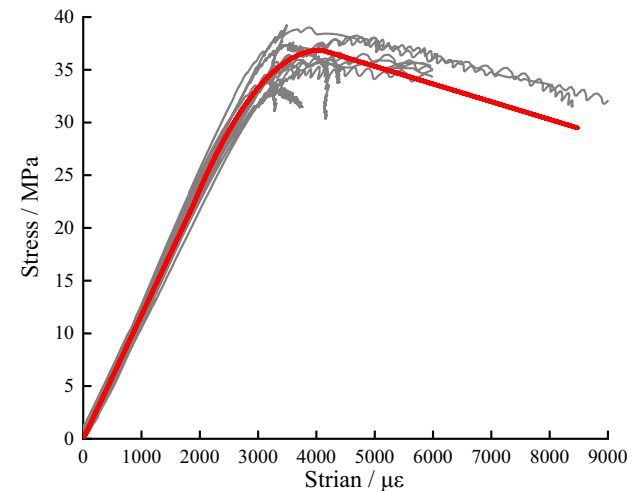
equation form was further simplified. The theoretical stress-strain curves of the CLTC and GLTC specimens in model III deviated more from the actual stress-strain curves in the elastoplastic range. Therefore, model III yielded a poor estimation for the performance of the CLTC and GLTC specimens in the elastoplastic range. In engineering applications, for correlation analysis, the stress-strain models I and II are recommended for the CLTC and GLTC, respectively. The stress-strain model III can be used for simple analyses of CLTC and GLTC during axial compression.

3.3. Analysis of the failure modes

As demonstrated, the axial compression performance of the CLTC and GLTC and their stress-strain relationships were quite different because of their lamination structures. Similarly, the axial compression failures of the CLTC and GLTC specimens were also substantially different. Fig. 8 shows the axial compression load-displacement curves of the CLTC and GLTC specimens. Because of greater compression strength of GLTC (Table 1), its bearing capacity was much higher than that of CLTC with the same cross-sectional area. Although the axial compression deformations of



(a) Model II of the CLTC specimens



(b) Model II of the GLTC specimens

Fig. 6. Model II of the axial compression stress-strain relationships of the CLTC and GLTC specimens.

both the CLTC and GLTC specimens can be divided into three ranges—the elastic range, elastoplastic range and plastic range—subtle differences still exist between the two specimen types.

In the elastic range, the vertical displacements of the CLTC and GLTC specimens increased linearly as the load of the test piece increased. Although the large material variability caused the load-displacement curves of the CLTC and GLTC specimens to overlap in the elastic phase, the elastic moduli of the GLTC specimens were still greater than those of the CLTC specimens. In this process, the wood cells of the GLTC specimens were compressed longitudinally. During the initial stage of loading in the CLTC specimens, the surface layers were mainly longitudinally compressed until the core layer was compacted to eliminate the gaps; then, the core layer worked to resist lateral compression. When the load was increased to the proportional limit, the specimen entered the elastoplastic range. As the load was subsequently increased, the slope of the load-displacement curve decreased continuously and the displacement rate began to increase. In the second half of the elastoplastic range, the cells inside the GLTC and CLTC specimens were compressed to an extent that they could not easily recover, which caused the cell walls to begin to buckle.

The GLTC and CLTC specimens appeared brittle near the end of the elastoplastic phase due to damage. After reaching the ultimate load, the specimens entered the plastic phase. As the load

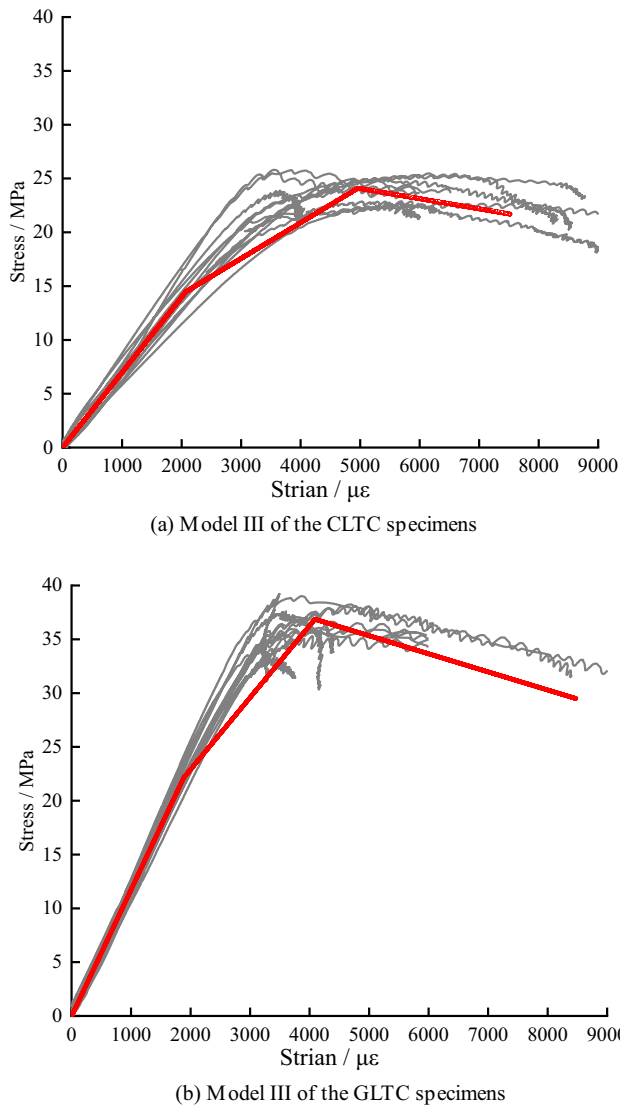


Fig. 7. Model III of the axial compression stress-strain relationships of the CLTC and GLTC specimens.

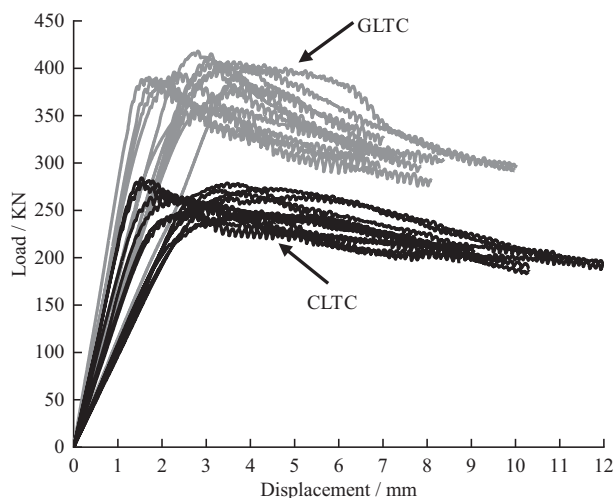


Fig. 8. Load-displacement curves.

increased, both the GLTC and CLTC specimens had large displacements. In this range, the slope of the load-displacement curve was negative. The longitudinal lamination around the narrow side of the CLTC specimen exhibited extensive shear and splitting damage. As shown in Fig. 9, the splitting damage was mostly present at the interface between the longitudinal layer and the transverse layer, and the longitudinal crack contained broken wood fibers. The cracks of the GLTC and CLTC specimens continued to develop as the load increased, and the cracks were connected to each other and through each side of the test piece.

As shown in Figs. 10 and 11, the GLTC and CLTC specimens mainly exhibited four failure modes: oblique shear failure (indicated by a circle), longitudinal cracking damage (represented by a rectangle), end-crushing damage (indicated by an ellipse), and cracking at the knot (indicated by a five-pointed star). The GLTC specimen mostly exhibited a single failure mode. However, the failure modes of the CLTC specimens were more complicated; almost all of the CLTC specimens exhibited mixed damage modes (Fig. 11). The failure modes of the narrow faces and wide faces of the GLTC specimen are the same, and the long and short column failure modes are basically similar (except for the end damage, the middle of the long column is prone to cracking). The failure modes of the narrow faces and wide faces of the CLTC specimens were substantially different, and the failure modes of the longer column specimens were generally more complex than those of the shorter column specimens. The oblique shear failures and the



Fig. 9. Splitting damage at the interface of the CLTC specimen.

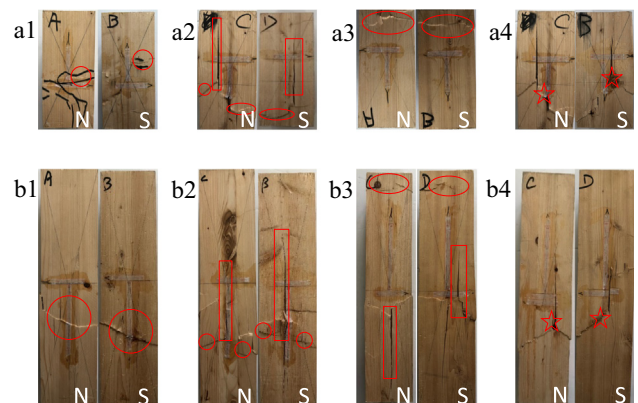


Fig. 10. Failure modes of the GLTC specimens.

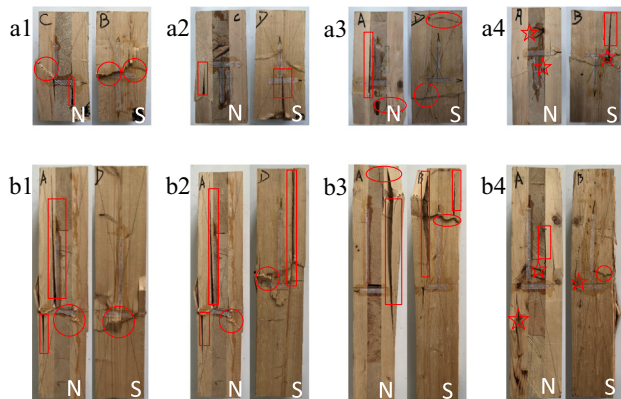


Fig. 11. Failure modes of the CLTC specimens.

end-crushing failures of the narrow faces of the CLTC specimens were mostly accompanied by longitudinal cracking damage at the gluing interface (Fig. 11a1, a3 N face and b1, b3 N face). The reason why this phenomenon occurred was that the adjacent layers were orthogonal to the increasing internal stress in the column, and the deformations of layers were different under axial compression; therefore, the different layers were sheared, and there was a gap between the substrates of the transverse layer. Eventually, the CLTC specimens exhibited complex destruction patterns. The failure mode of the wide face of the CLTC specimen was relatively simple. Notably, when the CLTC was axially compressed, it was prone to longitudinal cracking. Therefore, when a CLTC is used, it must be circumferentially reinforced.

4. Conclusions

In this work, axial compression experiments were conducted on two lengths of column specimens made from Canadian western hemlock CLT and control GLT. Through experimentation and analysis, the following conclusions were drawn:

1. The compression strength values of the GLT column specimens were significantly higher than those of the CLT column specimens. However, the CLT column specimens had enhanced ductility and energy absorption properties compared with the GLT column specimens due to the cross-lamination, which indicates that the CLT columns could have better connection performance than the GLT columns for certain applications.
2. Statistically, the column length did not have a significant effect on the compression MOE of the CLT and GLT column specimens, which indicates that the compression MOE obtained by the experiments is objective and can be used for the comparative analyses of the stress-strain models.
3. The failure process of the CLT and GLT column specimens under axial compression can be divided into 3 ranges: elastic, plastic and elastoplastic. Three stress-strain models were used for comparative analyses. For the GLT column specimens, model I was recommended, whereas for the CLT column specimens, model II was recommended. For simplicity, model III can also be used to analyze the axial compression behaviors of CLT and GLT.
4. Both the CLT and GLT column specimens exhibited four main failure modes: oblique shear failure, longitudinal splitting failure, end crushing and cracking, and wood joint cracking. However, the failure modes of the GLT column specimens were relatively simple, whereas the failure modes of the CLT column

specimens were more complicated due to their cross-lamination. Because the CLT columns are prone to longitudinal cracking during axial loading, a circumferential reinforcement is recommended.

5. Further studies are needed to examine the different connection behaviors of the CLT and control GLT columns and to develop the design properties of CLT columns for engineering applications.

Declaration of Competing Interest

None.

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